

SYSC 5108 Exam Study

Assignment 2, Question 6

Jesse Levine

Question

Consider training a logistic regression model using a loss close to least squared error (LSE). Assume $K = 2$, and define the differentiable loss through errors in probabilities:

$$J(\beta) = \sum_{i=1}^M \left[I_1(y^{(i)}) \left(1 - p_1(z^{(i)}) \right)^2 + I_2(y^{(i)}) \left(p_1(z^{(i)}) \right)^2 \right],$$

where

$$z^{(i)} = \beta^T x^{(i)}, \quad x_0 = 1.$$

Find the gradient of the loss function and compare it with the MLE gradient discussed in class.

Course and Textbook Cross-References

- **Chapter 3 slides, Logistic Regression Model:** for $K = 2$, logistic regression models the class probability using $p_1(x) = \sigma(z)$, where $z = \beta^T x$.
- **Chapter 3 slides, Alternative Notation:** for $y \in \{0, 1\}$, the negative log-likelihood loss has gradient

$$\nabla_{\beta} J_{\text{MLE}}(\beta) = - \sum_i \left(y^{(i)} - \sigma(z^{(i)}) \right) x^{(i)}.$$

- **Chapter 3 slides, Measuring Loss:** using negative log-likelihood is equivalent to cross-entropy for the model distribution.
- **Goodfellow, Bengio, and Courville, *Deep Learning*, Chapters 5 and 6:** logistic regression uses a sigmoid to map a linear score into $(0, 1)$, and maximum likelihood becomes minimization of negative log-likelihood, equivalently cross-entropy. The textbook also notes that mean squared error with sigmoid outputs can suffer from saturation and slow learning.

Step-by-Step Solution

1. Simplify the notation

For one sample i , define

$$p_i = p_1(z^{(i)}) = \sigma(z^{(i)}), \quad z^{(i)} = \beta^T x^{(i)}.$$

Also define the binary target

$$t_i = I_1(y^{(i)}).$$

Since there are only two classes,

$$I_2(y^{(i)}) = 1 - I_1(y^{(i)}) = 1 - t_i.$$

Therefore, the sample loss can be rewritten as

$$J_i(\beta) = t_i(1 - p_i)^2 + (1 - t_i)p_i^2.$$

This is exactly squared error on the predicted probability:

$$J_i(\beta) = (t_i - p_i)^2.$$

Check:

$$\begin{cases} t_i = 1 : & J_i = (1 - p_i)^2, \\ t_i = 0 : & J_i = p_i^2. \end{cases}$$

2. Differentiate the loss with respect to p_i

$$J_i = (t_i - p_i)^2.$$

Thus

$$\frac{\partial J_i}{\partial p_i} = 2(p_i - t_i).$$

Equivalently, using the original indicator notation:

$$\frac{\partial J_i}{\partial p_i} = -2I_1(y^{(i)})(1 - p_i) + 2I_2(y^{(i)})p_i.$$

3. Differentiate the sigmoid with respect to β

The sigmoid derivative is

$$\frac{d\sigma(z)}{dz} = \sigma(z)(1 - \sigma(z)).$$

Since $z^{(i)} = \beta^T x^{(i)}$,

$$\nabla_{\beta} z^{(i)} = x^{(i)}.$$

Therefore,

$$\nabla_{\beta} p_i = \frac{dp_i}{dz^{(i)}} \nabla_{\beta} z^{(i)} = p_i(1 - p_i)x^{(i)}.$$

4. Apply the chain rule

For one sample,

$$\begin{aligned} \nabla_{\beta} J_i(\beta) &= \frac{\partial J_i}{\partial p_i} \nabla_{\beta} p_i \\ &= 2(p_i - t_i)p_i(1 - p_i)x^{(i)}. \end{aligned}$$

Summing over all M samples gives

$$\boxed{\nabla_{\beta} J(\beta) = 2 \sum_{i=1}^M (p_i - t_i) p_i (1 - p_i) x^{(i)}}.$$

Using $t_i = I_1(y^{(i)})$ and $p_i = p_1(z^{(i)})$, this is

$$\nabla_{\beta} J(\beta) = 2 \sum_{i=1}^M \left(p_1(z^{(i)}) - I_1(y^{(i)}) \right) p_1(z^{(i)}) \left(1 - p_1(z^{(i)}) \right) x^{(i)}.$$

In the assignment's original indicator notation:

$$\nabla_{\beta} J(\beta) = 2 \sum_{i=1}^M \left[-I_1(y^{(i)})(1 - p_i) + I_2(y^{(i)})p_i \right] p_i(1 - p_i)x^{(i)}.$$

Comparison With MLE

For binary logistic regression, the MLE or negative-log-likelihood loss is

$$J_{\text{MLE}}(\beta) = - \sum_{i=1}^M [t_i \log(p_i) + (1 - t_i) \log(1 - p_i)].$$

Its gradient is

$$\nabla_{\beta} J_{\text{MLE}}(\beta) = \sum_{i=1}^M (p_i - t_i)x^{(i)} = - \sum_{i=1}^M (t_i - p_i)x^{(i)}.$$

Now compare the two gradients:

$$\nabla_{\beta} J_{\text{LSE-prob}}(\beta) = 2 \sum_i (p_i - t_i) \underbrace{p_i(1 - p_i)}_{\text{extra sigmoid derivative}} x^{(i)}$$

while

$$\nabla_{\beta} J_{\text{MLE}}(\beta) = \sum_i (p_i - t_i)x^{(i)}.$$

Interpretation

- The LSE-on-probabilities gradient has an extra factor $p_i(1 - p_i)$, and $p_i(1 - p_i) \leq 0.25$.
- When the sigmoid saturates, $p_i \approx 0$ or $p_i \approx 1$, this extra factor becomes very small.
- This is a serious problem when the model is *confident but wrong*. For example, if $t_i = 1$ but $p_i \approx 0$, then

$$\nabla_{\beta} J_{\text{MLE},i} \approx -x^{(i)},$$

which gives a strong corrective update, but

$$\nabla_{\beta} J_{\text{LSE-prob},i} \approx -2p_i(1 - p_i)x^{(i)} \approx 0,$$

which can make learning very slow.

- This is why MLE/cross-entropy is preferred for logistic regression and sigmoid output units: the log-likelihood cancels the sigmoid derivative in a way that avoids this extra saturation factor.

Final Answer

$$\nabla_{\beta} J(\beta) = 2 \sum_{i=1}^M \left(p_1(z^{(i)}) - I_1(y^{(i)}) \right) p_1(z^{(i)}) \left(1 - p_1(z^{(i)}) \right) x^{(i)}$$

Compared with MLE,

$$\nabla_{\beta} J_{\text{MLE}}(\beta) = \sum_{i=1}^M \left(p_1(z^{(i)}) - I_1(y^{(i)}) \right) x^{(i)}.$$

The LSE-style probability loss therefore has the same prediction-error term $(p_i - t_i)$, but it also has the extra sigmoid-derivative factor $p_i(1 - p_i)$, which can cause vanishing gradients for saturated sigmoid outputs.

Exam Note

If the loss is written with a leading factor $\frac{1}{2}$, then the factor 2 in the LSE gradient disappears:

$$\frac{1}{2}(t_i - p_i)^2 \quad \Rightarrow \quad \nabla_{\beta} J_i = (p_i - t_i)p_i(1 - p_i)x^{(i)}.$$

The key comparison with MLE does not change: the LSE probability loss still has the extra $p_i(1 - p_i)$ factor.